SIMATS ENGINEERING

CONTINOUS INTERNAL ASSESSMENT- CONCEPT MAPPING

COURSE NAME: DATABASE MANAGEMENT SYSTEMS
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Questions:05 **Total marks:100**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I R Charulatha / 192426003 3 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Database Management System CO1AT2	R. charulatha 1924260033 CSA0507

1 EER Model - Generalization, Specialization and Aggregation.

The Enhanced Entity - Relationship (EER) model extends the basic ER model by adding advanced concepts such as generalization, specialization, and aggregation.

* Generalization: combines two or more lower-level entities into a higher-level superclass based on their common properties.

Example - student, Faculty and Staff can be generalized into person.

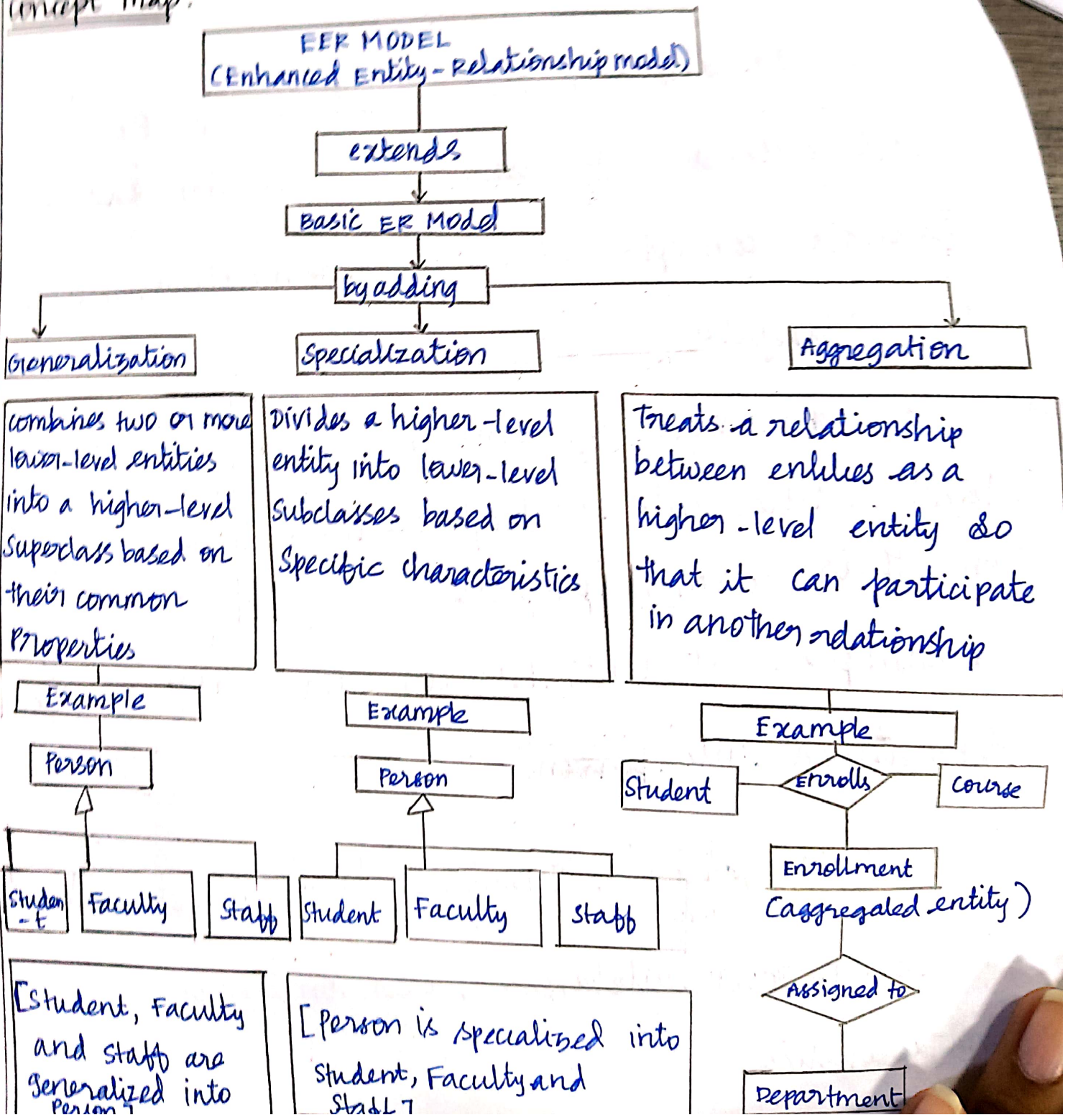
* Specialization : Divides a higher-level entity into lower-level subclasses based on specific characteristics.

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Example: Person can be specialized into ... and staff

Aggregation: Treats a relationship between ... a higher-level entity so that it can participate ... another relationship.

Concept map:

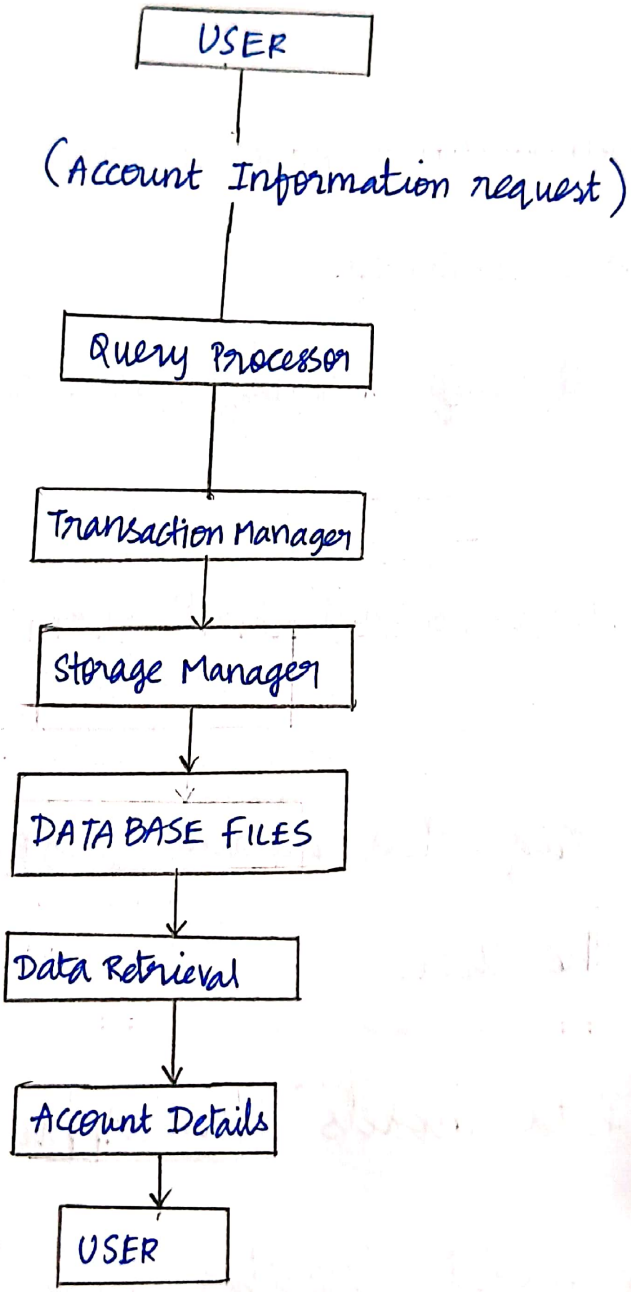


MS Software
when a user requ...
passes through ...
before the req...

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DBMS Software Architecture - online Banking:

When a user requests account information, the request passes through different components of the DBMS before the required data is returned.



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Explanation:

1. The user sends a request for account information.

2. The query processor interprets and executes the query.

3. The transaction manager manages the transaction and maintains consistency.

4. The storage manager accesses the required stored data.

5. The data is retrieved from the database files.

6. The requested account information is returned to the user.

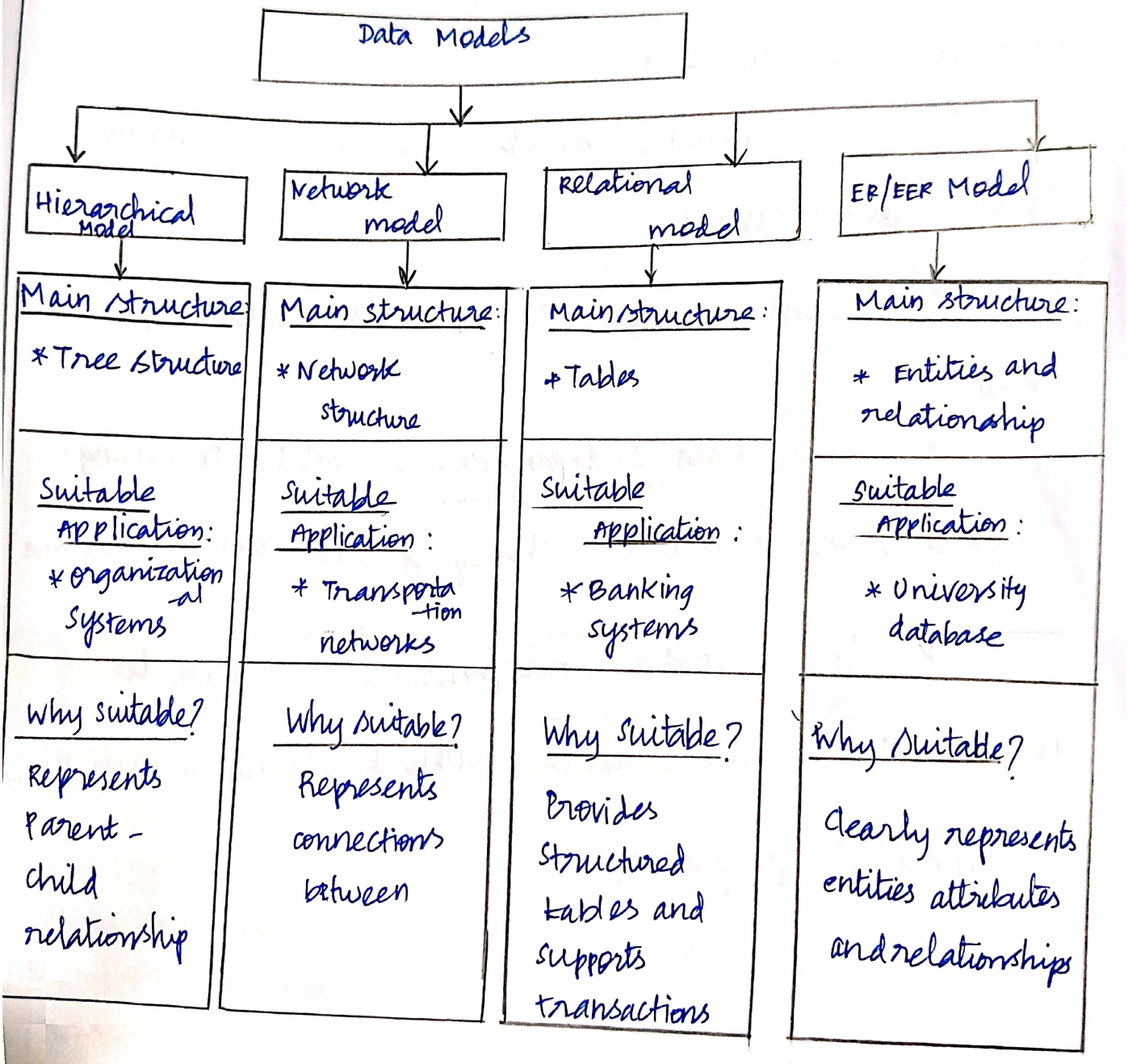
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3 Different Data models and Applications:

A data model describes how data is organized stored and related in a database. Different models are suitable for different applications.

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ta Model	Main structure
Hierarchical Model	Tree structure
Network Model	Network / graph structure
Relational Model	Tables
ER/EER Model	Entities & relationships



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4 Schema, Instance and Data Independence:

A schema is the overall structure or design of a database, while an instance is the actual data stored in the database at a particular time.

* Schema: structure / design of the database.

* Instance: Actual data present in the database at a particular moment.

* Logical schema: Describes the logical structure of the entire database.

* Physical schema: Describes how data is physically stored.

* Physical Logical Data Independence: Ability to change physical storage without changing the logical schema.

* Logical Data Independence: Ability to change the logical schema without changing user view or application programs.

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Schema, Instance and Data Independance

Schema

* Structure or design of the database
* Shows how data is organised

Instance

* Actual data in the database at a given time
* Changes frequently

Logical schema

* Describes the logical structure of the database
* (What data is stored and how it is related)

Physical schema

* Describes how data is physically stored in database
* (Files, indexes, storage, structures)

Logical Data Independence

* Change logical schema without affecting user views or application programs

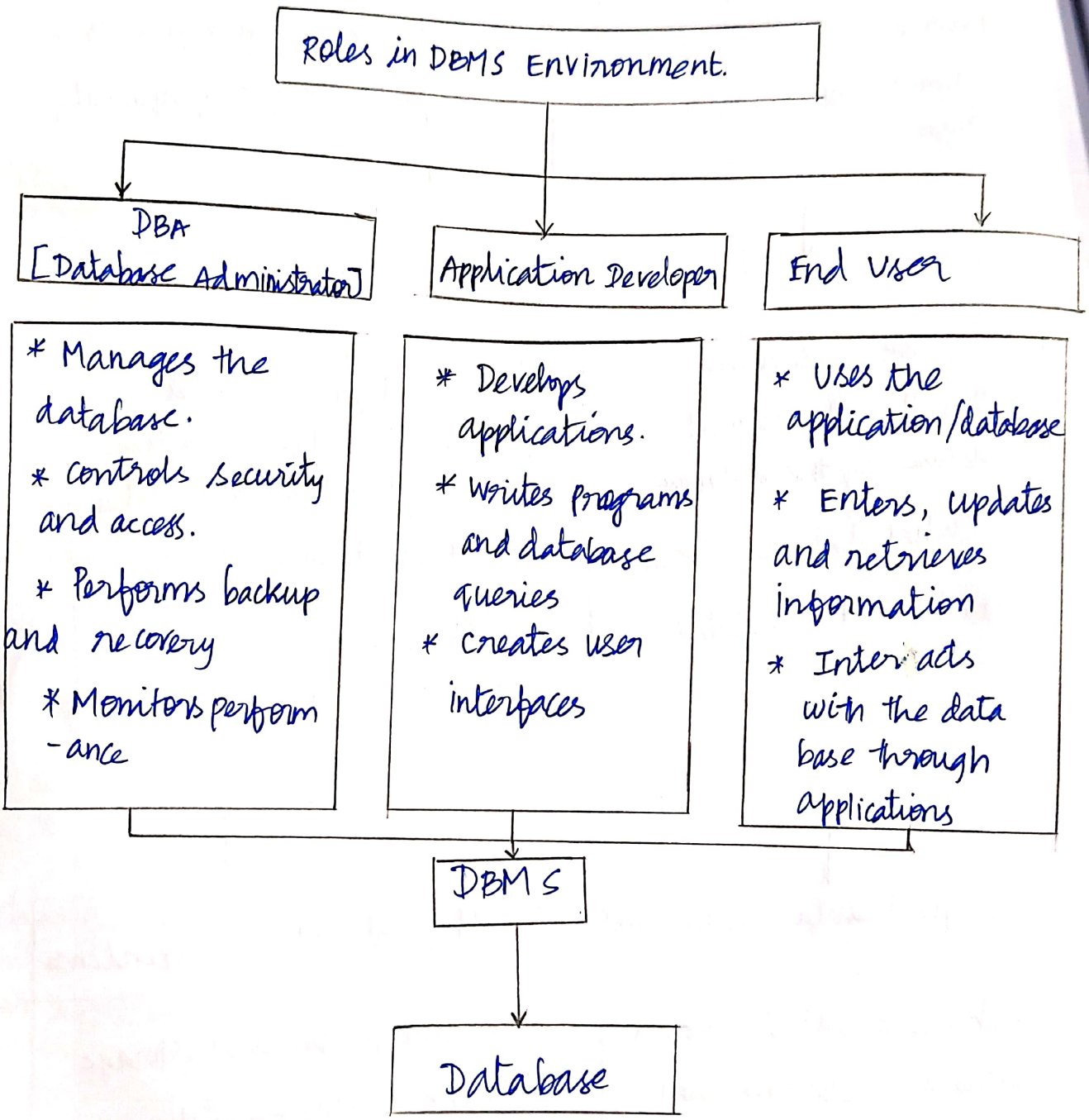
Physical Data Independence

* Change physical storage without affecting the logical schema.

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5 Roles in a DBMS Environment:

A DBMS environment involves different users who perform different activities. The three important roles are Database Administrator (DBA), End User, and Application Developer.



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...sibilities :

1) DBA

* Manages the database.
* Performs backup and recovery.
* Monitors database performance.

2) Application Developer

* Develops applications that interact with the database.
* Creates user interfaces
* Writes programs and database queries.

3) End User

* Uses the application / database
* Enters, Updates and retrieves information.
* Interacts with the database through applications

Overall interaction

End user
↓
Application Developer's application
↓
DBMS
↓
DBA
↓
Database

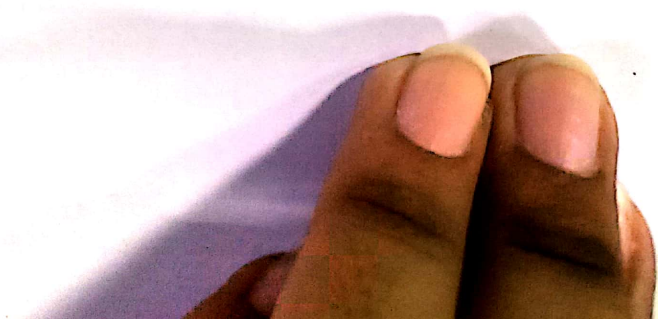
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ANNEXURE
CONCEPT MAPPING

QUESTIONS

Q.No	Question	Bloom's Level
1	A university database contains Persons, Students, Faculty, and Staff, where specialized entities inherit common properties from a general entity. Explain how the EER model extends the basic ER model using generalization, specialization, and aggregation. Construct a concept map illustrating these concepts and their relationships.	BL2 – Understand
2	An online banking application receives a user's request to retrieve account information. Construct a visual concept map explaining the flow of the request through the DBMS software architecture, including the query processor, transaction manager, storage manager, database files, and data retrieval process.	BL2 – Understand
3	Different organizations use different data models according to their requirements. Construct a concept map connecting Hierarchical, Network, Relational, and ER/EER models with suitable real-world applications such as organizational systems, transportation networks, banking systems, and university databases. Explain the suitability of each model.	BL2 – Understand
4	A university database undergoes changes in its logical structure and physical storage without changing how users access the data. Construct a concept map explaining schema, instance, logical schema, physical schema, logical data independence, and physical data independence, and show the relationship among these concepts.	BL2 – Understand
5	An organization uses a DBMS to manage its business data. Construct a concept map showing the roles of the Database Administrator (DBA), End User, and Application Developer. Include their major responsibilities and explain how they interact within the overall DBMS environment.	BL1 – Remember

92/100



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